

BUREAU OF INDIAN STANDARDS

INDIAN STANDARD: IS 4984 : 2016

HIGH DENSITY POLYETHYLENE (HDPE) PIPES FOR WATER SUPPLY — SPECIFICATION

Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi 110002

1 SCOPE

Specifies requirements for high density polyethylene pipes intended for buried and above-ground water supply systems.

Table 1: Material Properties (Clause 5.1)

Sl No.	Property	Permissible Bound	Test Method
1	Base Polymer Density at 27°C	940 to 958 kg/m ³	IS 7328
2	Melt Flow Index (190°C / 5 kg)	0.20 to 1.10 g/10min	IS 2530
3	Carbon Black Content	2.0 to 3.0 %	IS 2530

Table 4: Hydrostatic Strength (Clause 8.1)

Sl No.	Test Duration	Test Temperature	Hoop Stress (PE 100)
1	100-Hour Acceptance Test	20°C	12.4 MPa
2	165-Hour Type Test	80°C	5.4 MPa